

Snake lemma

Euna Park

Let R be a unital ring. All modules are left R -modules. All groupoids are small, with discrete object and arrow sets.

Definition 1 (Translation groupoid). For a homomorphism $f : M \rightarrow N$, let $G_f = N//M$ have objects $y \in N$ and arrows $y \xrightarrow{m} y + f(m)$ for $m \in M$, with composition given by addition of labels. Write $X_f = |\mathbf{N}(G_f)|$, based at 0. Here $\mathbf{N}$ denotes the nerve and $|-|$ geometric realisation.

Proposition 2. *Naturally in f , there are isomorphisms $\pi_1(X_f, 0) \cong \ker f$ and $\pi_0 X_f \cong \operatorname{coker} f$, and $\pi_j(X_f, 0) = 0$ for $j \geq 2$.*

Proof. The components of G_f are the cosets of $f(M)$, and $\operatorname{Aut}_{G_f}(y) = \ker f$. Each component is equivalent to the one-object groupoid $\ker f$, whose classifying space is $K(\ker f, 1)$. A commutative square $(u, v) : f \rightarrow g$ induces a functor $G_f \rightarrow G_g$ given by v on objects and u on labels. ■

Definition 3. For a functor of groupoids $P : G \rightarrow H$, the *strict fibre* $P^{-1}(h)$ consists of the objects over h and the arrows over id_h . The functor P is an *isofibration* if, for every $x \in G$ and $h \in H$, every arrow $u : P(x) \rightarrow h$ lifts to an arrow $v : x \rightarrow y$.

For $q : E \rightarrow D$ and $d_0 \in D$, fix the path model

$$\operatorname{hofib}_{d_0}(q) = \{(e, \lambda) \mid \lambda : [0, 1] \rightarrow D, \lambda(0) = q(e), \lambda(1) = d_0\}.$$

Lemma 4. *If $P : G \rightarrow H$ is an isofibration and $F = P^{-1}(h)$, then*

$$|\mathbf{N}(F)| \longrightarrow |\mathbf{N}(G)| \longrightarrow |\mathbf{N}(H)|$$

is homotopy fibre sequence. With basepoint $x_0 \in F$, its boundary sends $u : h \rightarrow h$ to the component of the endpoint of a lift starting at x_0 .

Proof. The comma groupoid $P \downarrow h$ has objects $(x, u : P(x) \rightarrow h)$ and arrows $v : (x, u) \rightarrow (y, w)$ with $wP(v) = u$. Postcomposition with any arrow of H is an isomorphism of comma groupoids. Quillen's Theorem B therefore identifies $|\mathbf{N}(P \downarrow h)|$ with the homotopy fibre [1, §1, Theorem B].

The inclusion $F \rightarrow P \downarrow h$, $x \mapsto (x, \operatorname{id}_h)$, is fully faithful. A lift $v : x \rightarrow y$ of u gives $(x, u) \xrightarrow{\sim} (y, \operatorname{id}_h)$, proving essential surjectivity. The inclusion induces the comparison map, given by constant paths. For (x_0, u) , the isomorphism $(x_0, u) \xrightarrow{\sim} (y, \operatorname{id}_h)$ gives the boundary formula. ■

Since $\pi_2|N(H)| = 0$, the long exact sequence of homotopy groups gives

$$\begin{aligned} 1 \longrightarrow \pi_1|N(F)| \longrightarrow \pi_1|N(G)| \longrightarrow \pi_1|N(H)| \\ \xrightarrow{\partial} \pi_0|N(F)| \longrightarrow \pi_0|N(G)| \longrightarrow \pi_0|N(H)|, \end{aligned} \quad (1)$$

where basepoints are omitted and exactness at π_0 is understood in pointed sets.

Example 5. Take $R = \mathbb{Z}$ and $m \geq 2$. Reduction of labels modulo m gives an isofibration

$$G_{\text{id}:\mathbb{Z} \rightarrow \mathbb{Z}} \longrightarrow G_{0:\mathbb{Z}/m \rightarrow 0}.$$

Its strict fibre has objects $\mathbb{Z}$ and arrows $n \rightarrow n + mk$ for $k \in \mathbb{Z}$. The total groupoid and each fibre component have contractible classifying spaces. Thus the sequence is modelled by

$$(\mathbb{Z}/m)_{\text{disc}} \longrightarrow E(\mathbb{Z}/m) \longrightarrow B(\mathbb{Z}/m),$$

where $E(\mathbb{Z}/m)$ is a contractible free $\mathbb{Z}/m$ -space and $B(\mathbb{Z}/m)$ its quotient. Lifting $\bar{r}$ to 0 $\xrightarrow{r} r$ gives $\partial(\bar{r}) = \bar{r}$.

Theorem 6 (Snake lemma). *A commutative diagram of left R -modules with short exact rows*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C & \longrightarrow & 0 \\ & & \alpha \downarrow & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & A' & \xrightarrow{i'} & B' & \xrightarrow{p'} & C' & \longrightarrow & 0 \end{array}$$

determines a natural exact sequence

$$\begin{aligned} 0 \longrightarrow \ker \alpha &\xrightarrow{i} \ker \beta \xrightarrow{p} \ker \gamma \\ &\xrightarrow{\delta} \text{coker } \alpha \xrightarrow{\bar{i}'} \text{coker } \beta \xrightarrow{\bar{p}'} \text{coker } \gamma \longrightarrow 0. \end{aligned}$$

For $c \in \ker \gamma$, choose $b \in B$ with $p(b) = c$ and write $\beta(b) = i'(a')$. Then $\delta(c) = [a']$.

Proof. The right-hand square induces a functor $P : G_\beta \rightarrow G_\gamma$ given by p' on objects and p on labels. Exactness of the rows identifies the strict fibre of P over 0 with G_α . Since p is surjective, P is an isofibration. Lemma 4 and Proposition 2 identify (1) with the six-term sequence of kernels and cokernels. The map $\bar{p}'$ is surjective since p' is.

The lift $0 \xrightarrow{b} \beta(b)$ ends at $i'(a')$, giving $\delta(c) = [a']$. Changing b by $i(a)$ changes a' by $\alpha(a)$. Linear combinations of lifts show that δ is R -linear, so the sequence is exact as a sequence of R -modules. The relations $p(b) = c$ and $\beta(b) = i'(a')$ are preserved by morphisms of diagrams, so δ is natural. $\blacksquare$

References

- [1] D. Quillen, Higher algebraic K -theory: I, in Algebraic K -theory I, Lecture Notes in Mathematics 341, Springer, 1973, pp. 85–147.